

FAST NUCES

Better Split - Final Report

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Better Split - Final Project Report

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1 Introduction

Better Split is a desktop application designed to help groups of friends, roommates, and colleagues easily manage and split shared expenses. The application provides an intuitive, user-friendly, and secure platform where users can track spending, divide costs fairly, settle debts efficiently, and maintain clear financial records within groups.

This project was developed as part of the Fundamental Software Engineering course, where we applied agile methodologies and real-world software engineering practices. Throughout the development process, we engaged in sprint planning, user story creation, design documentation, architecture planning, testing, and task management using tools like Trello.

2 User Stories (All Sprints)

2.1 Sprint 1

2.1.1 Module 1: User Authentication

Main Story:
As a User, I want to be able to sign up and log in to my account securely so that I can access my expenses privately.

Sub-Stories:

- As a User, I want to register using an email and password.
- As a User, I want to log in using my registered credentials.

Acceptance Criteria:
I know I am done when I can successfully create an account and log in using my credentials.

2.1.2 Module 2: Group Creation

Main Story:
As a User, I want to create groups with friends for different expenses so that I can easily manage shared costs.

Sub-Stories:

- As a User, I want to name my group.
- As a User, I want to invite members to join the group.
- As a User, I want to edit or delete a group if needed.

Acceptance Criteria:
I know I am done when I can create a group and add participants.

2.1.3 Module 3: Adding Expenses

Main Story:
As a User, I want to add expenses to a group and assign them to members so that the system can automatically calculate shared costs.

Sub-Stories:

- As a User, I want to select a group before adding an expense.
- As a User, I want to enter details like amount, description, and payer.
- As a User, I want to split the expense equally or manually adjust shares.

Acceptance Criteria:
I know I am done when I can enter expenses and see the system distribute costs among group members.

2.2 Sprint 2

2.2.1 Module 4: Group Creation

Main Story:
As a user, I want to create and invite members to a group to manage shared expenses easily.

Sub-Stories:

- As a user, I want to name the group for easy identification.
- As a user, I want to view a list of all members in a group.

Acceptance Criteria:
I know I am done when I can successfully create a group, add participants, and view the list of group members.

2.2.2 Module 5: Expense Splitting

Main Story:
As a user, I want to log an expense and split it fairly so that everyone knows what they owe.

Sub-Stories:

- As a user, I want to add details like amount and payer while creating an expense.
- As a user, I want to edit or delete an expense if a mistake is made.

Acceptance Criteria:
I know I am done when I can successfully add, edit, or delete an expense and the system fairly splits and categorizes the costs among group members.

2.2.3 Module 6: Settle Up

Main Story:
As a user, I want to see who I owe and pay them with as few transactions as possible.

Sub-Stories:

- As a user, I want to view a summary of all debts and credits.
- As a user, I want to mark a debt as "Settled" once payment is done.
- As a user, I want to receive a confirmation after settling up.

Acceptance Criteria:
I know I am done when I can view my balances, mark debts as settled, and receive confirmation that my payment has been recorded.

2.3 Sprint 3

2.3.1 Module 7: Data Export

Main Story:
As a user, I want to export my expense data so that I can keep personal records or share them externally.

Sub-Stories:

- As a user, I want to export the expenses of a specific group to a CSV file.
- As a user, I want the exported data to be formatted clearly, including the names, amounts, and categories of the expenses.

Acceptance Criteria:

- I know I am done when I can export group expenses into a readable file format (CSV) with the required details.
- I know I am done when I can select a date range for the exported data.
- I know I am done when the export is well-organized, including the necessary expense details like names, amounts, and categories.

2.3.2 Module 8: Data Import

Main Story:
As a user, I want to import expense data into the system so that I can easily integrate external records.

Sub-Stories:

- As a user, I want to import expense data from a CSV or Excel file.
- As a user, I want the imported data to be automatically added to the correct group.
- As a user, I want to be notified if any imported data is missing or incorrect.

Acceptance Criteria:

- I know I am done when I can import a CSV or Excel file with expense data.
- I know I am done when the imported data is correctly categorized and added to the appropriate group.
- I know I am done when the system alerts me if there are errors in the imported data (e.g., missing information or invalid format).

2.3.3 Module 9: Multiple Groups

Main Story:
As a user, I want to be able to create and manage multiple groups so that I can track different sets of shared expenses.

Sub-Stories:

- As a user, I want to create multiple groups for different sets of people or events (e.g., a family group and a roommate group).
- As a user, I want to switch between multiple groups without losing my data.
- As a user, I want to view the summary of each group separately, showing debts and credits for each group individually.

Acceptance Criteria:

- I know I am done when I can create multiple groups and manage them independently.
- I know I am done when I can switch between groups without data loss or confusion.
- I know I am done when I can see a separate summary for each group, including the balance and debts.

2.3.4 Module 10: Testing

Main Story:
As a developer, I want to ensure the system works correctly by performing comprehensive testing on new features and existing ones.

Sub-Stories:

- As a tester, I want to verify the correct import and export of data formats (CSV).
- As a tester, I want to ensure that multiple groups function independently and data doesn't overlap between them.
- As a tester, I want to test edge cases such as invalid file formats during import or large datasets.

Acceptance Criteria:

- I know I am done when all testing scenarios pass for the Data Export, Data Import, and Multiple Groups features.

- I know I am done when there are no errors in the import/export of data files.
- I know I am done when the system works smoothly when switching between multiple groups.

3 Design

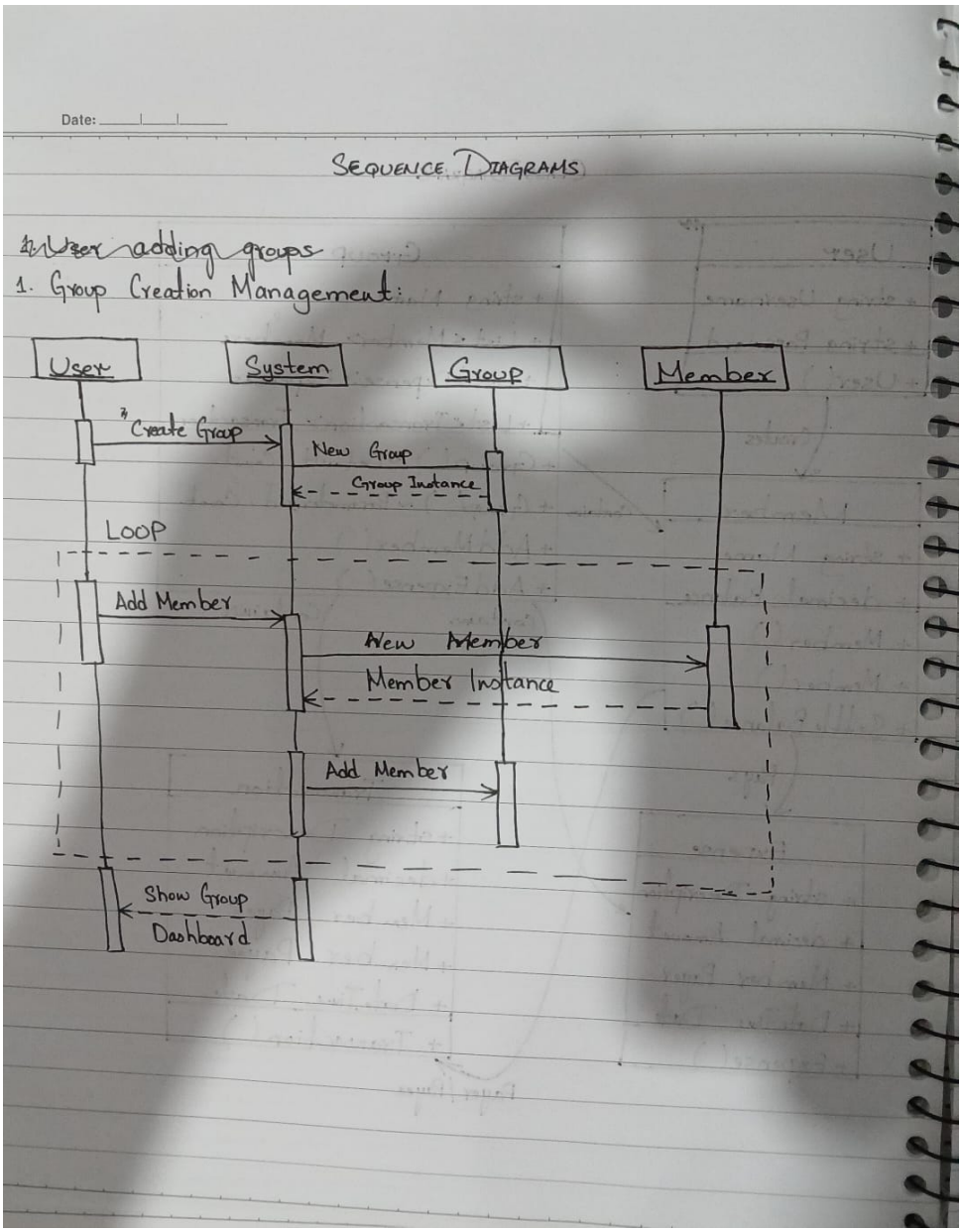


Figure 1: System Design Overview

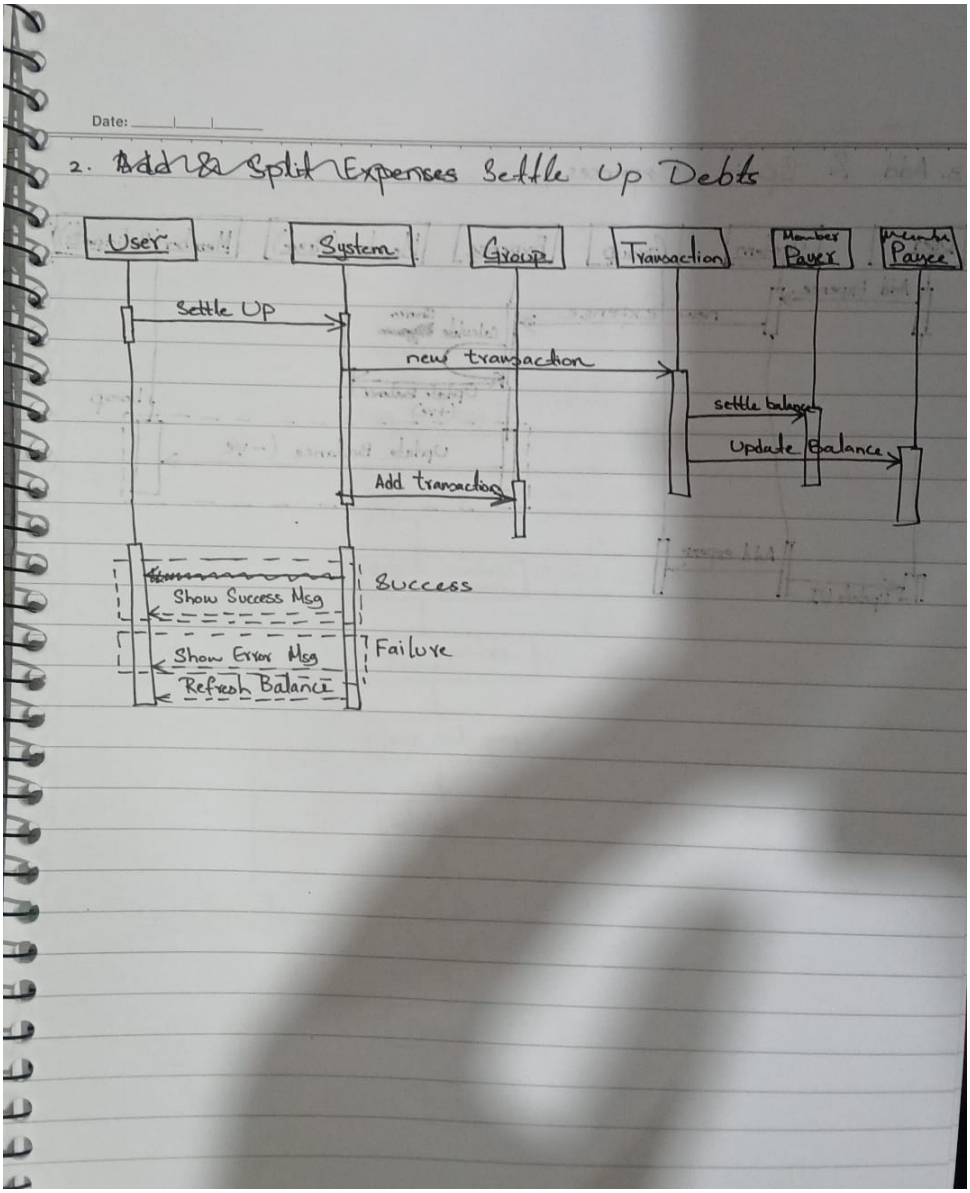


Figure 2: System Design Overview

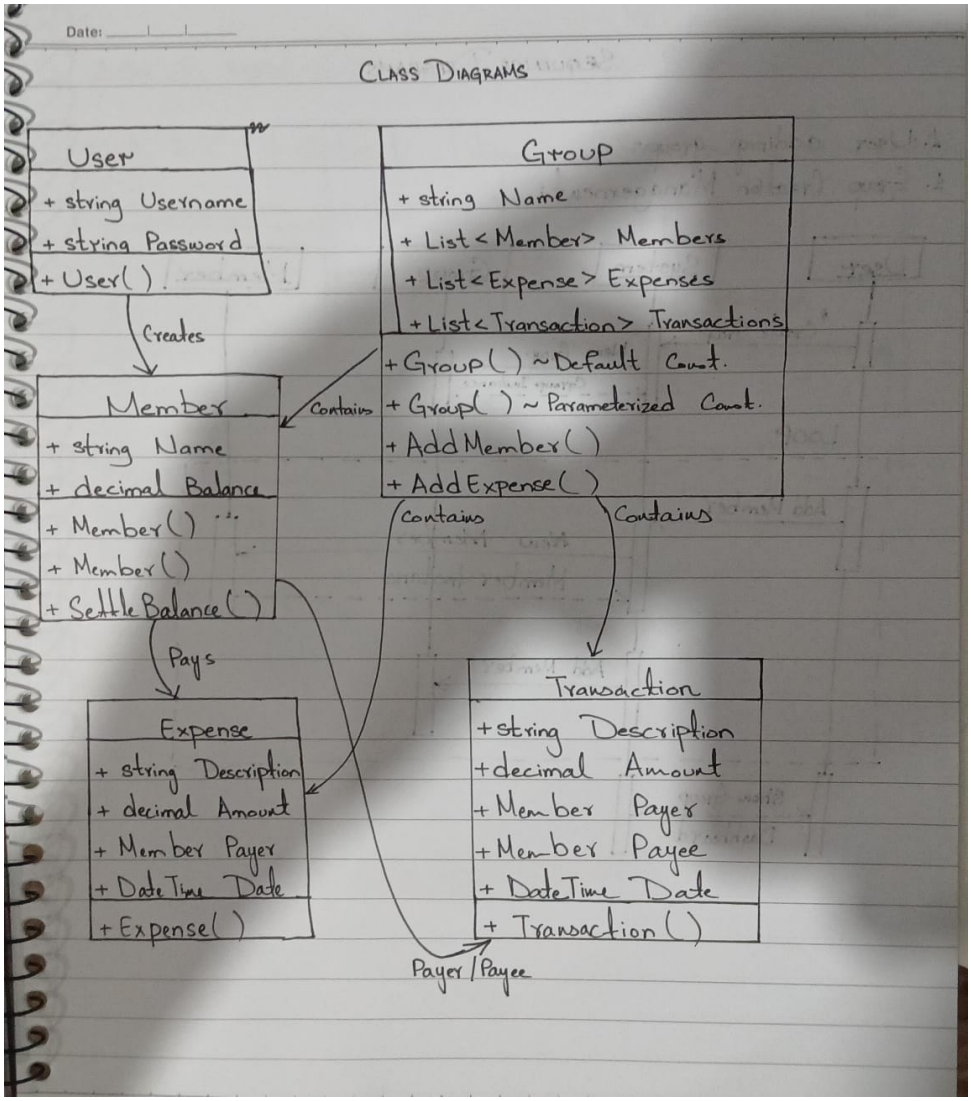


Figure 3: System Design Overview

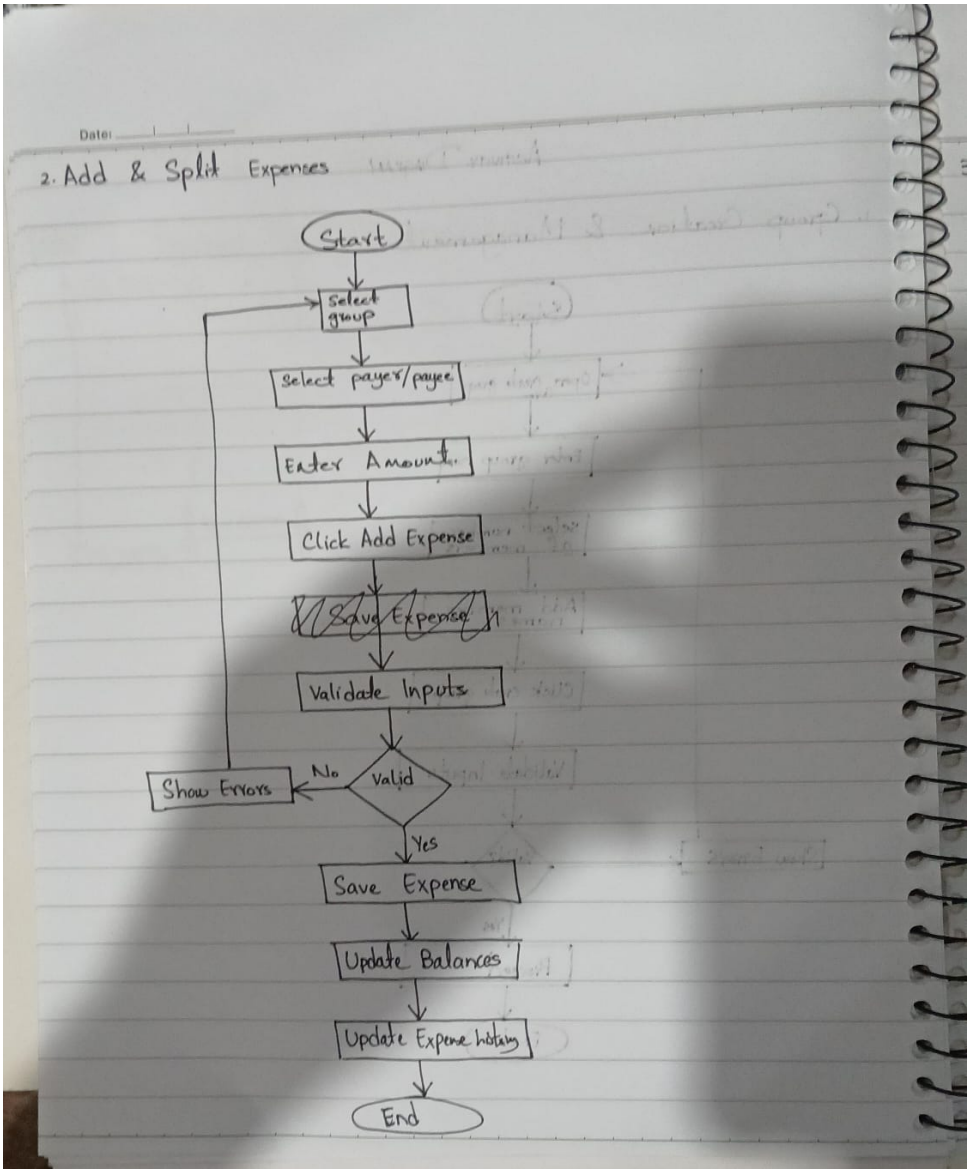


Figure 4: System Design Overview

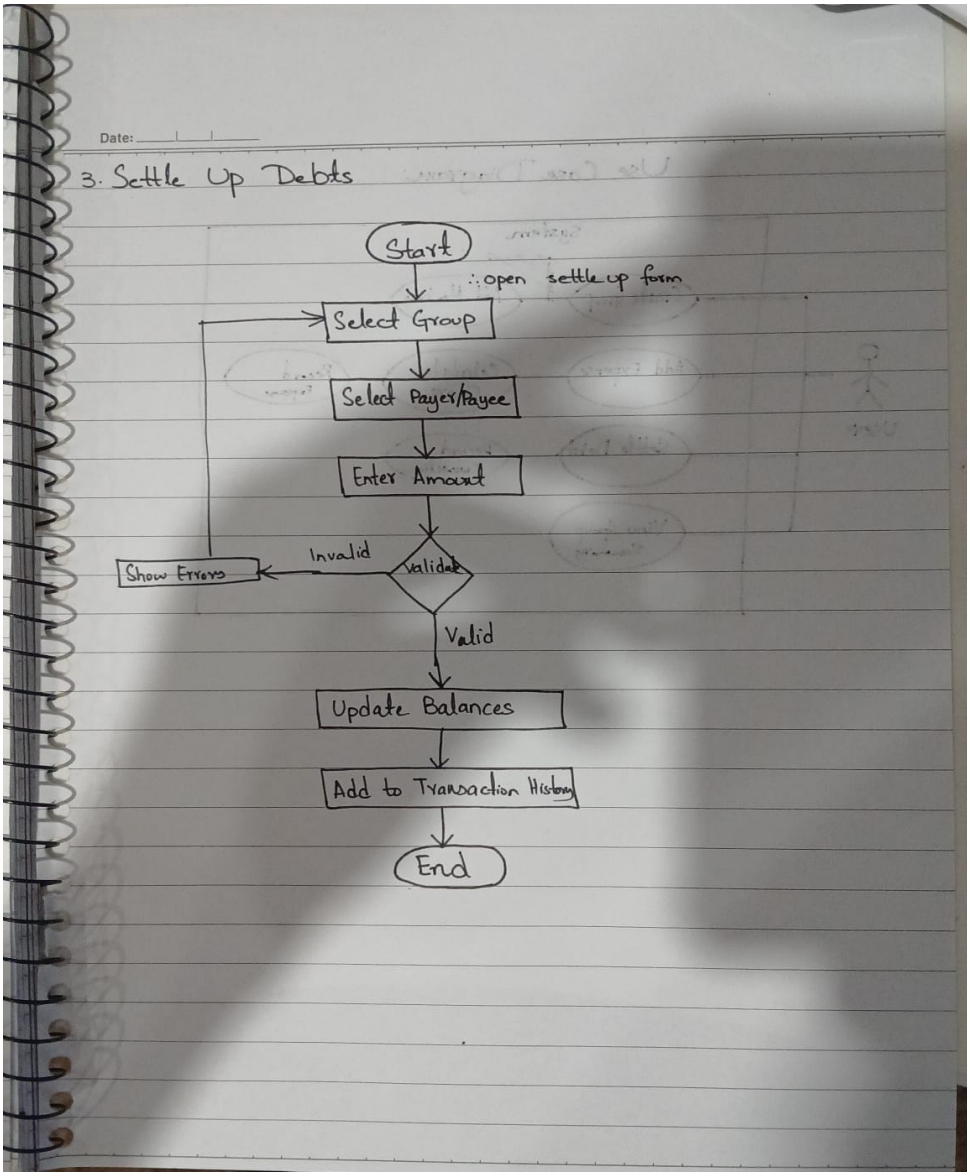


Figure 5: System Design Overview

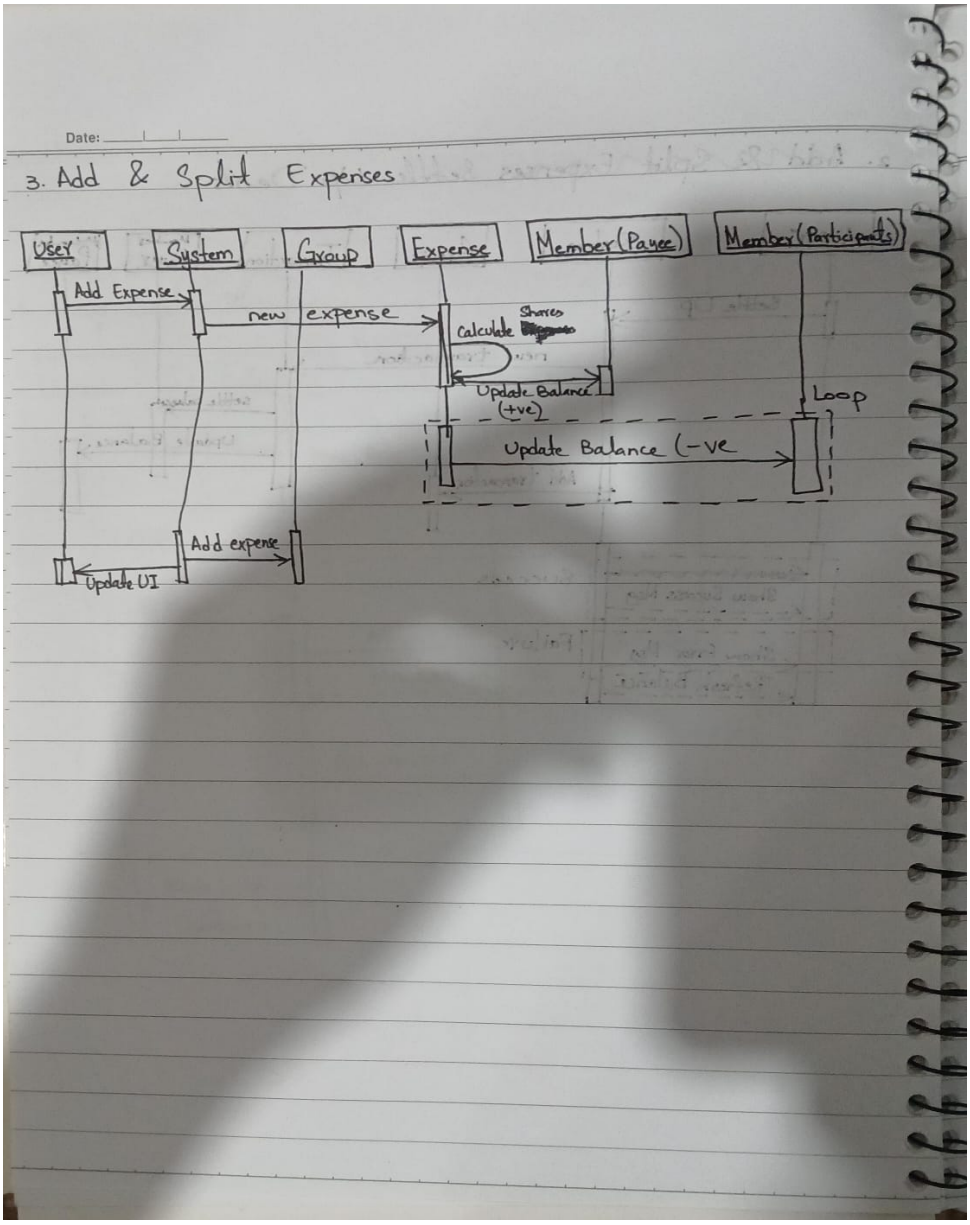


Figure 6: System Design Overview

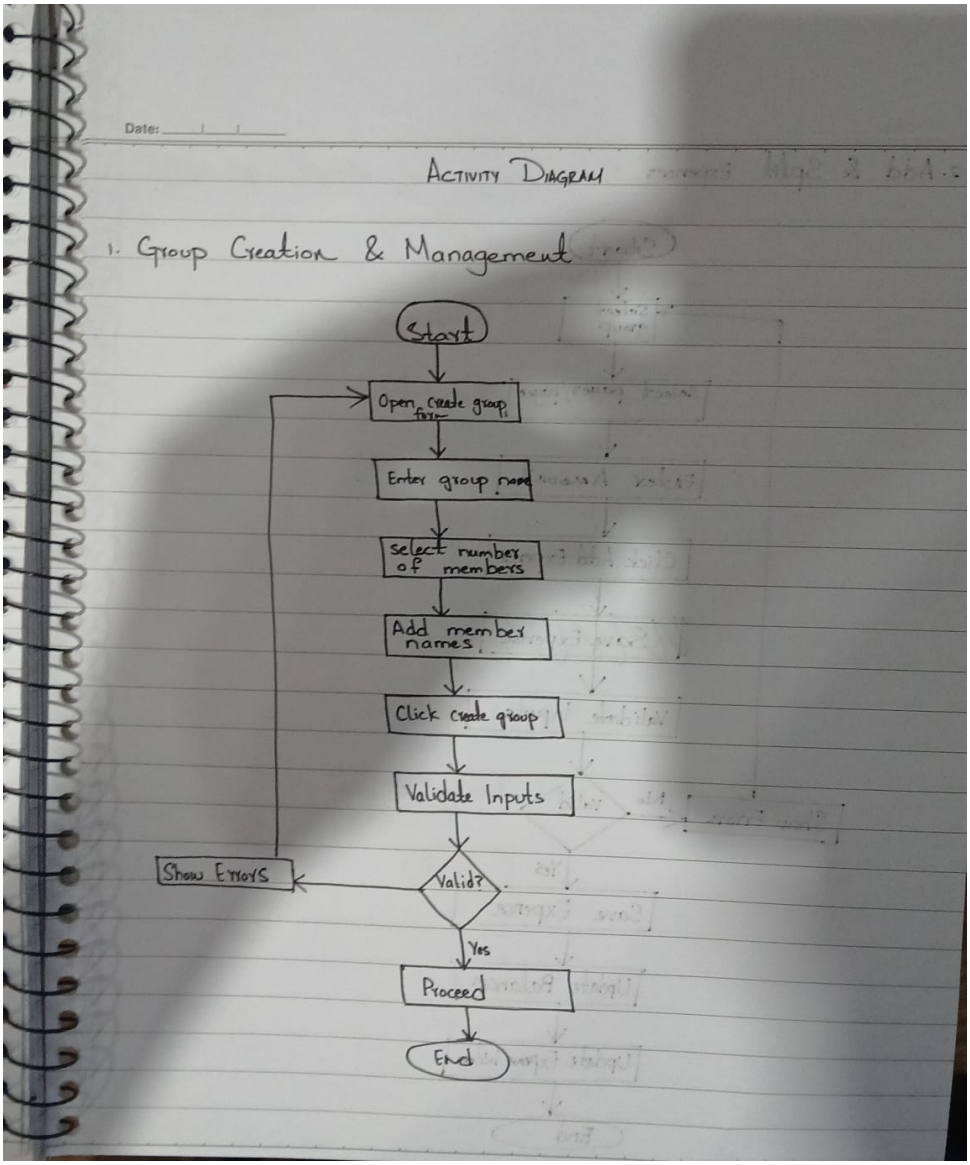


Figure 7: System Design Overview

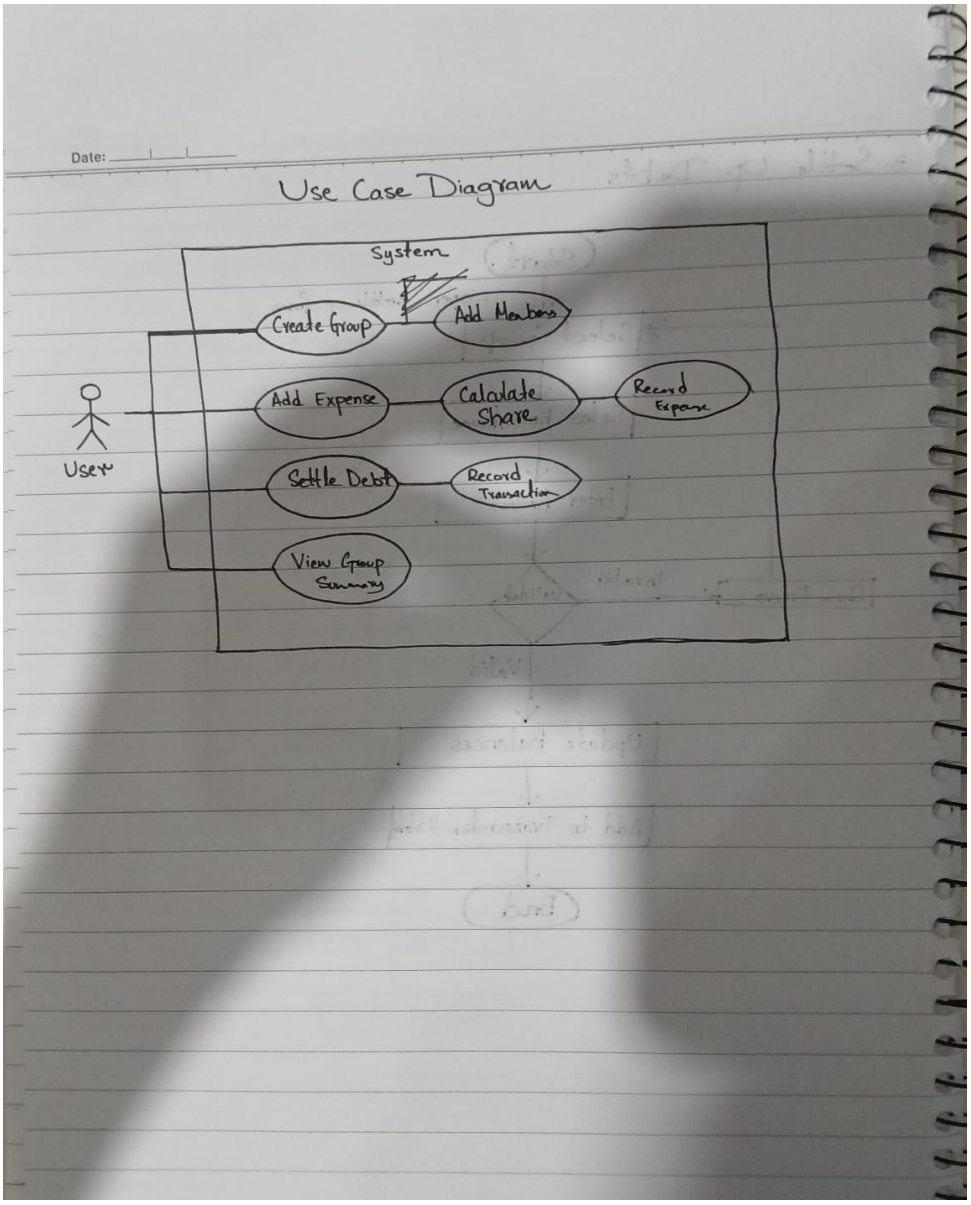


Figure 8: System Design Overview

4 Architecture

BetterSplit is designed using the MVC (Model-View-Controller) architecture.

4.1 Model

The Model handles all the data and business logic. It stores information about users, groups, expenses, and debts. It performs calculations for splitting expenses, tracking debts, and settling balances.

4.2 View

The View is responsible for displaying information to the users. It provides the user interface screens for creating groups, adding expenses, viewing balances, and settling up.

It updates automatically when any data changes in the Model.

4.3 Controller

The Controller manages the interaction between the View and the Model. It processes user actions like submitting a new expense, creating a group, or settling a debt. It updates the Model with new data and refreshes the View to show the updated state.

4.4 Flow

The user interacts with the View (e.g., adds an expense). The Controller captures the input, processes it, and updates the Model. The Model updates the data and notifies the View to reflect the changes. This separation helps manage the application's complexity by dividing it into clear layers.

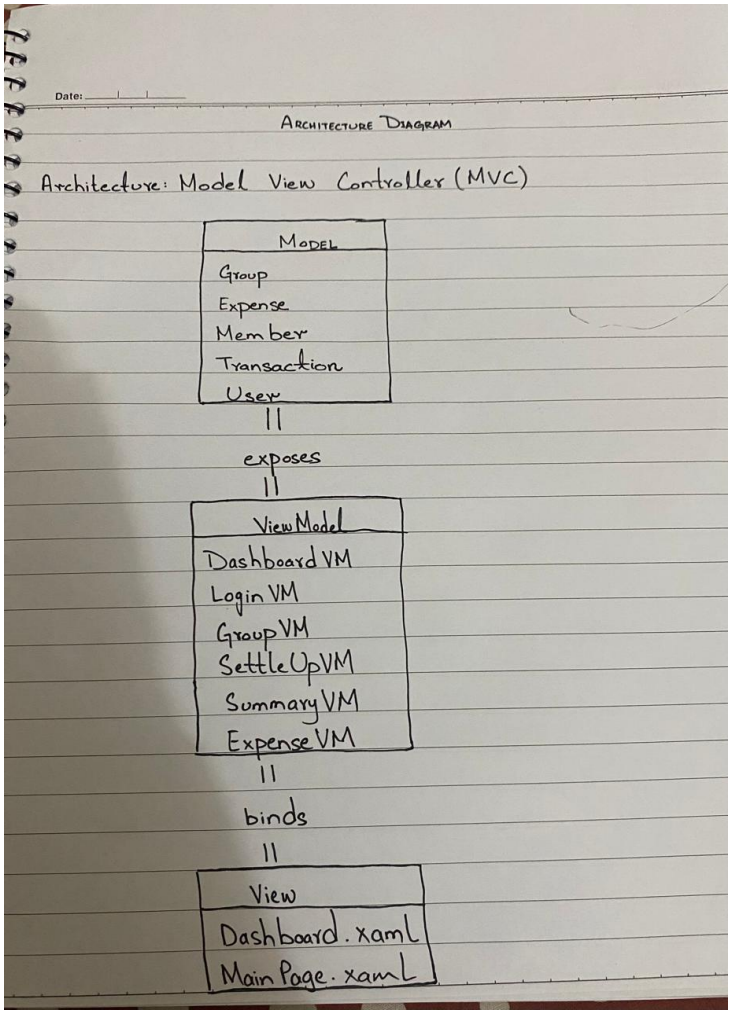


Figure 9: System Architecture Overview

5 Actual Implementation

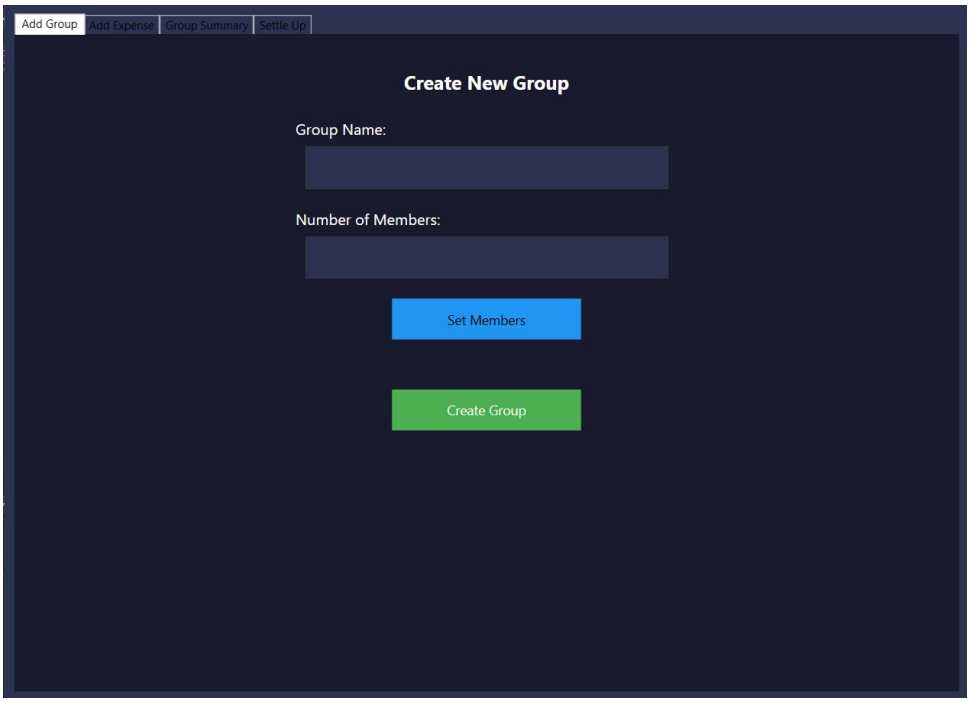


Figure 10: System Implementation Overview

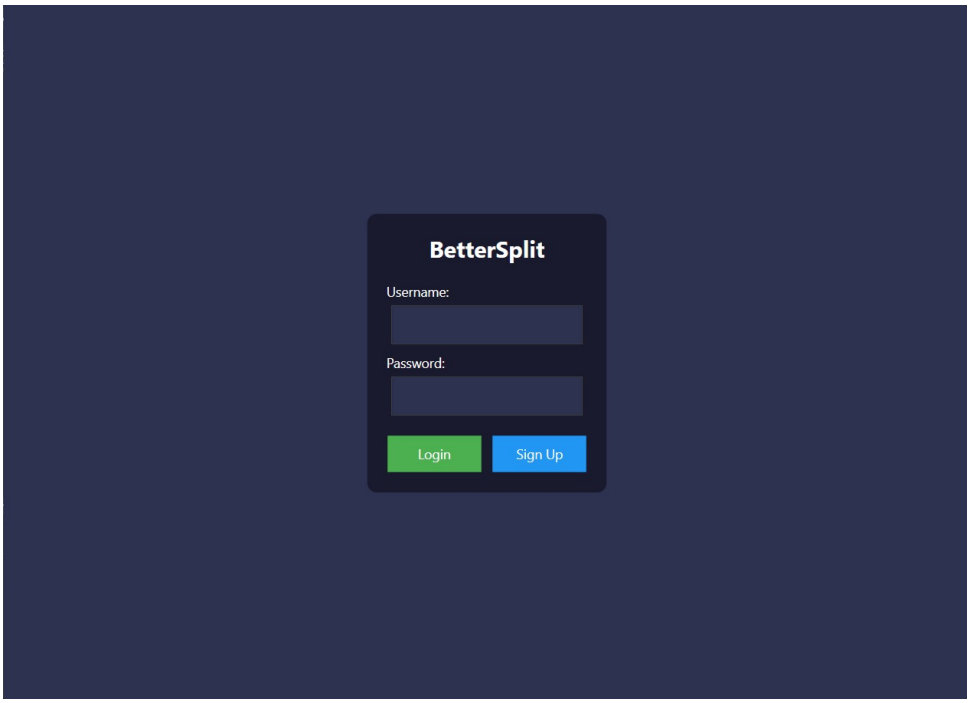


Figure 11: System Implementation Overview

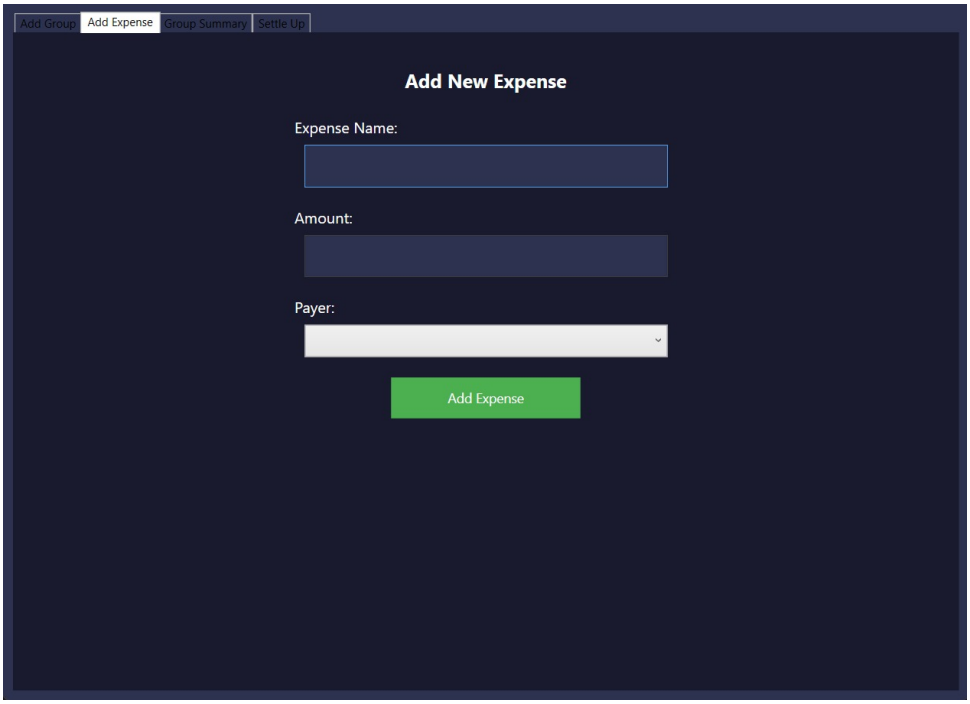


Figure 12: System Implementation Overview

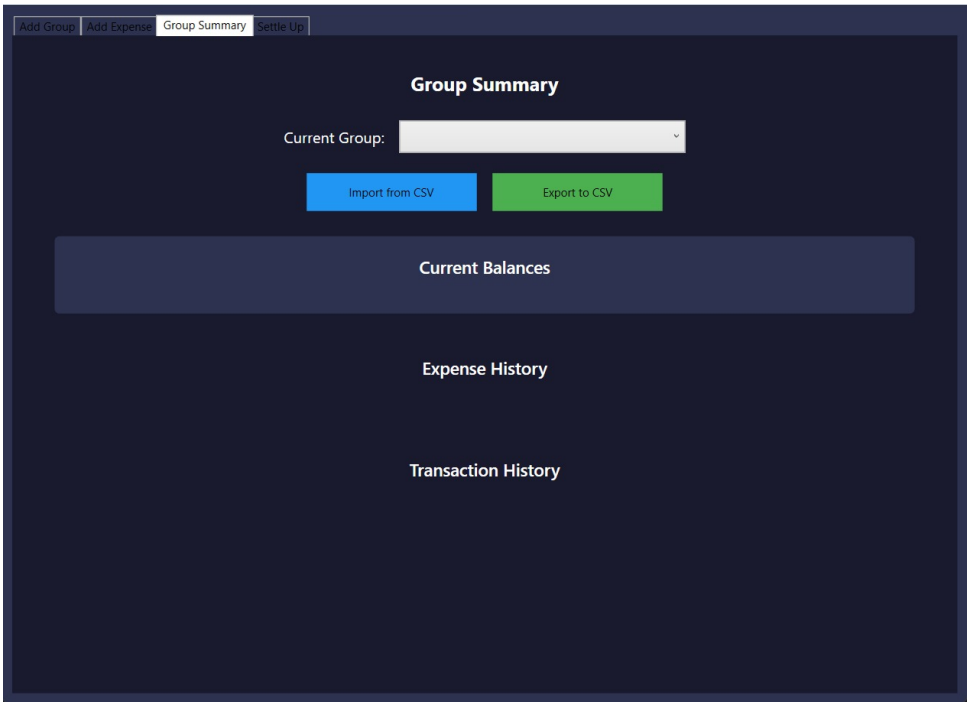


Figure 13: System Implementation Overview

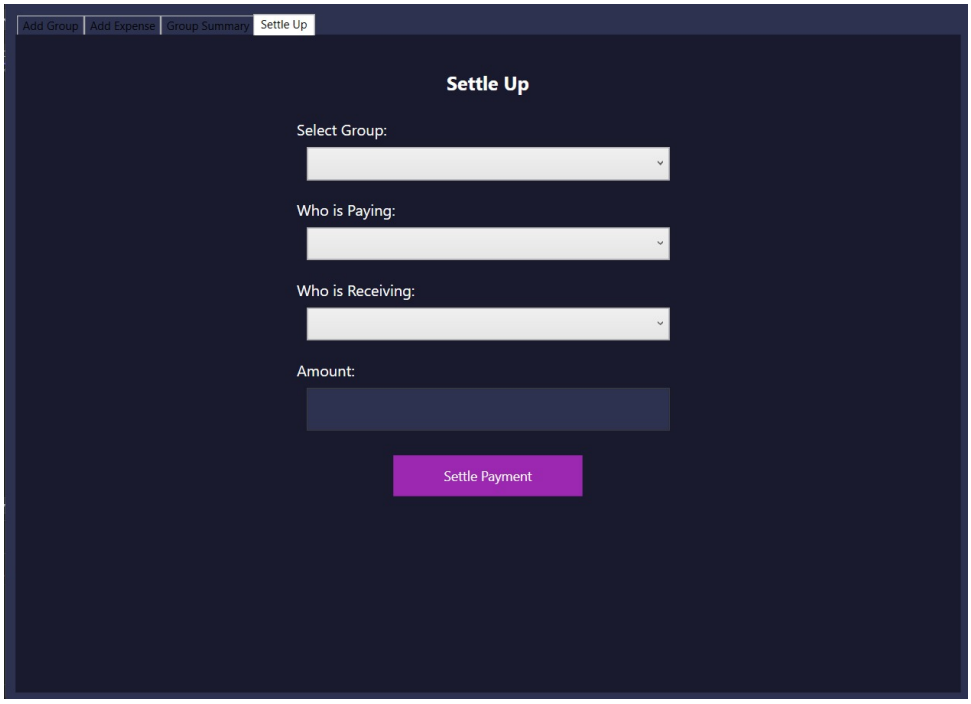


Figure 14: System Implementation Overview

6 Product Burn Down Chart

Week	Planned Remaining Points	Actual Remaining Points (Example)
0	37	37
1	33	35
2	29	32
3	25	28
4	21	23
5	17	20
6	13	16
7	9	11
8	5	5
9	0	0

Figure 15: Product Burn Down Chart

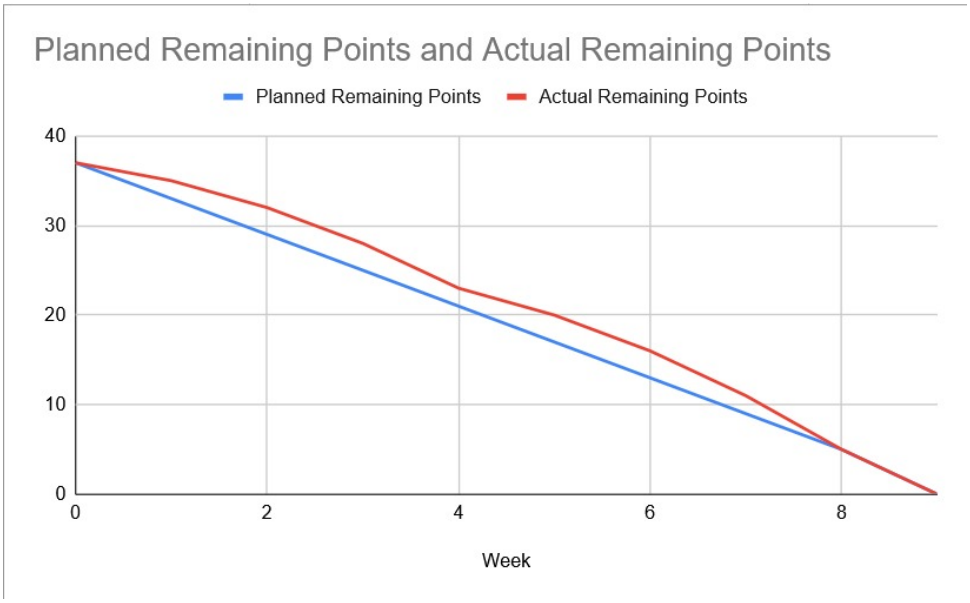


Figure 16: Product Burn Down Chart

7 Trello Board Screenshot

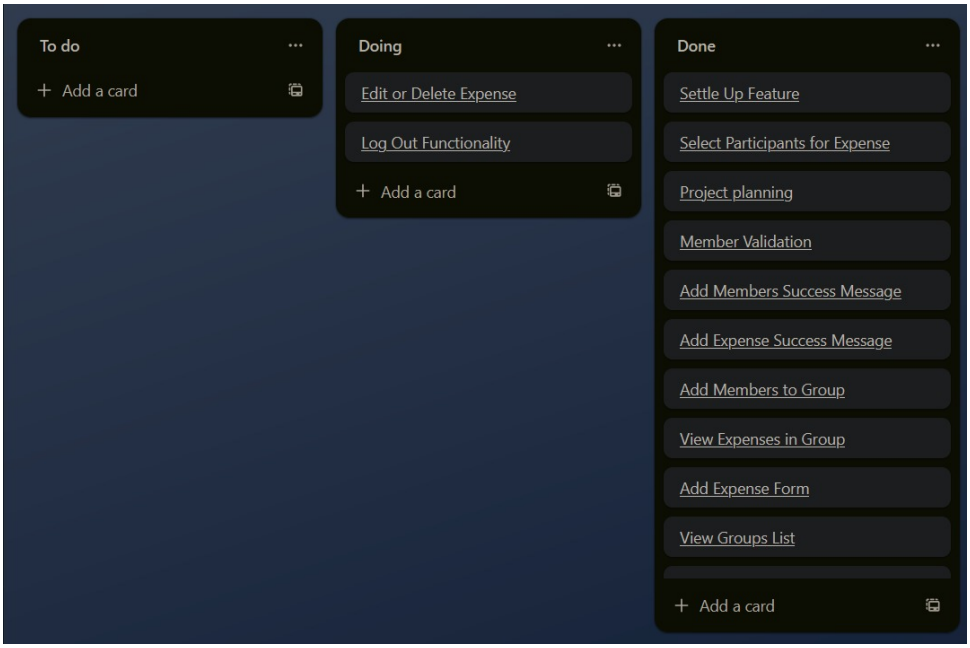


Figure 17: Trello Board Used for Task Management

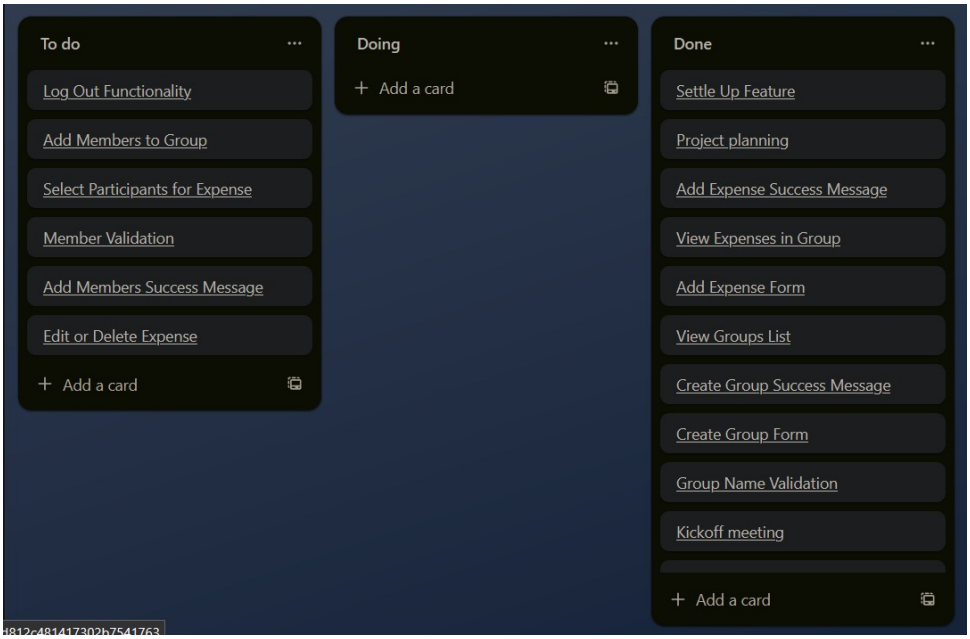


Figure 18: Trello Board Used for Task Management

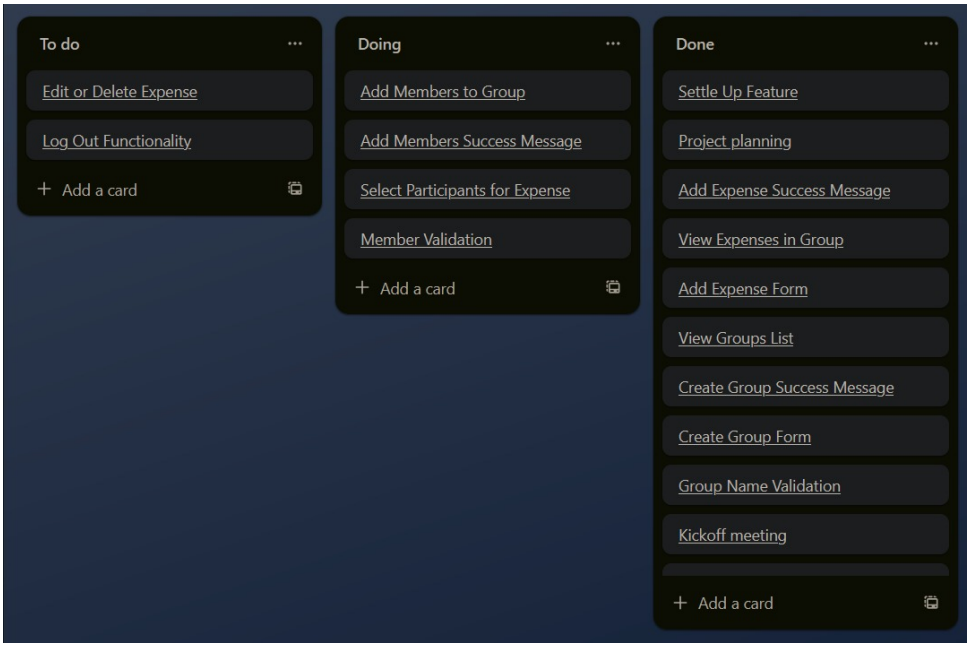


Figure 19: Trello Board Used for Task Management

8 Boundary Value Analysis Testing

Test Case	Username	Password	Expected Result	Reason
1	ab	Password1!	Fail	Username too short (2 characters)
2	abc	Password1!	Pass	Username at minimum valid length (3 characters)
3	abcdefghijklmno	Password1!	Pass	Username at maximum valid length (15 characters)
4	abcdefghijklmnop	Password1!	Fail	Username exceeds maximum length (16 characters)
5	abc	pass1!	Fail	Password missing uppercase letter
6	abc	PASSWORD1!	Fail	Password missing lowercase letter
7	abc	Password!	Fail	Password missing number
8	abc	Password1	Fail	Password missing special character
9	abc	Pass1!	Pass	Password at minimum valid length (6 characters)
10	abc	PasswordPassword1!	Pass	Password at maximum valid length (20 characters)
11	abc	PasswordPassword1!X	Fail	Password exceeds maximum length (21 characters)
12	abc user	Password1!	Fail	Username contains invalid space
13	user_123	Password1!	Pass	Username with underscore allowed
14	user$123	Password1!	Fail	Username contains invalid character $

Figure 20: Boundary Value Testing

9 Work Division Between Groups

Throughout the development of BetterSplit, all three members — Muneeb, Okasha, and Kabeer — contributed equally to the project. Each member was responsible for different yet equally important aspects of the project to ensure a smooth and collaborative workflow.

9.1 Muneeb

Handled core backend development, focusing on implementing the Model and Controller components. Developed the expense calculation logic and debt settlement functionalities. Participated actively in sprint planning, user story writing, and testing.

9.2 Okasha

Focused on frontend development, designing the user interface and building the View components. Implemented screens for group management, adding expenses, and displaying balances. Contributed to documentation tasks, design planning, and sprint retrospectives.

9.3 Kabeer

Took responsibility for system integration and ensuring smooth communication between frontend and backend (linking View, Controller, and Model). Worked on project management tasks, maintaining the Trello board and preparing burn down charts. Led the preparation of the final report, user story breakdowns, and testing documentation.

Each member was involved in:

- Designing and reviewing the system architecture.
- Contributing to the overall UI/UX design decisions.
- Writing and reviewing user stories across all sprints.
- Testing the application (manual and boundary value testing).

- Weekly group meetings and sprint reviews to ensure project consistency.

Thus, the workload was distributed evenly among all three members, allowing the project to be completed collaboratively and successfully.

10 Learning From Project

Working on the BetterSplit project provided valuable practical experience in applying software engineering principles and real-world project management techniques. Some key learnings from the project are:

- **Understanding of MVC Architecture:** We gained hands-on experience in implementing the Model-View-Controller architecture, understanding how to properly separate the business logic, user interface, and control flow to build scalable and maintainable applications.
- **Agile Methodologies:** Through sprint planning, user story creation, and sprint retrospectives, we learned how to break down a large project into manageable tasks and deliver work iteratively and incrementally.
- **Task Division and Team Collaboration:** Dividing work among group members while maintaining regular communication helped us understand the importance of clear role assignment, regular meetings, and teamwork in software projects.
- **Full Software Development Lifecycle (SDLC) Experience:** From initial project proposal to final testing and documentation, we experienced each phase of the software development lifecycle.
- **Testing and Quality Assurance:** Writing and executing boundary value analysis tests helped us appreciate the importance of systematic testing in ensuring the application is reliable and bug-free.
- **Version Control and Project Management Tools:** Using tools like Trello for task tracking and burn down charts for progress monitoring taught us how professional teams organize and monitor project progress.
- **Problem-Solving and Adaptability:** During the project, we encountered unexpected technical and design challenges. Solving these collaboratively improved our problem-solving skills and ability to adapt under pressure.
- **UI/UX Importance:** We realized how important a simple, intuitive, and clean user interface is for the success of any application, especially when managing sensitive data like expenses.

Overall, the BetterSplit project helped bridge the gap between theoretical knowledge and practical application, preparing us for larger software engineering projects in the future.